

Late 2003 witnessed the birth of the then-third-fastest supercomputer, affectionately termed the Big Mac. The cluster-based supercomputer was birthed by a research team at Virginia Tech University, headed by Srinidhi Varadarajan—the Director of the Terascale Computing Facility at Virginia Tech and an Assistant Professor in the Department of Computer Science. Big Mac was strung together from 1,100 dual-processor Power Mac G5s, and was the first supercomputer with Mac inside. Big Mac was also affordable to digest, costing a relatively modest US \$5.2 million. To compare: Japan's Earth Simulator (then the second fastest supercomputer) cost an estimated US \$350 million to US \$400 million. While the Big Mac is capable of performing about 12 trillion operations a second (teraflops), the Earth Simulator operates at about 35 teraflops.

The budget supercomputer was declared the third-fastest, today it takes up the seventh position. Virginia Tech's next bargain supercomputer, due sometime in 2006, is all set to aim at for 50 teraflops, if status quo is maintained; this would allow Virginia to reclaim the third spot.

TREND > THE PC IN YOUR LIVING ROOM

One PC To Entertain Them

As a general trend, technology is migrating from servers and desktops into our living rooms. We have all been amused by the idea of our refrigerators e-mailing a grocery list to our local store, or our cell phones wirelessly programming our microwave ovens; these are intriguing concepts but not solutions in themselves. What is more interesting is the concept of home entertainment and the trends in that space.

We are of course talking about PCs which masquerade as consumer devices in your living room. Take the HP z545 Digital Entertainment Center as an example. It looks like a stereo component and will be right at home in a rack-mounted setup, next to a TV set. Innocuous on the outside, but take a close look: fold-down panels on the front hide two USB 2.0 ports, a CF, an SD, and a Memory Stick slot, you will also find RCA inputs, S-Video in, a Firewire port, a dual-layer DVD-RW drive and an alpha-numeric display. Turn it around and the back of the unit is overwhelmed with output ports: full 7.1-channel audio out (RCA and optical), a VGA port, S-video out, DVI video out, S/PDIF, composite video out, Component video out, a FireWire port, and four USB 2.0 ports. The most interesting piece to the equipment is the "Personal Media" drive. Which is a 7,200 RPM USB 2.0 portable hard drive with a docking station build inside the z545—the portable disk will let you carry 160 GB of data to a friend's or a relative's. That data can be TV shows, movies, music or photos. This is adjunct to the 200 GB hard disk that lives inside the unit...

But a nifty black box does not home entertainment make. Also starring are TV sets in more flavours than ice cream: the LCD TV (the largest from Sharp stands at 65-inch), the Wi-Fi LCD TV such as Philips' 23-inch Streamium 23PF9976i, which has built-in 802.11g for streaming video and audio files, the Plasma TV (which can be as



This Samsung Plasma TV, like all Plasma TVs, come in sizes upto 63-inch

large as 63-inch), the Projection TV which are lighter and use less power than a Plasma TV, also frequenting 63-inch. A supporting cast featuring a 4.1-, 5.1- or 7.1-channel speaker setup, projectors with in-built DVD and 2.1-channel audio support and a comfortable couch to sprawl on.

TECHNOLOGY > INTERNET TELEPHONY

VoIP—Waiting For Voice Quality

There is little doubt that VoIP is good technology. In a nutshell it allows international voice calls to be made over the Internet. This in turn brings down the cost of the call borne by a user. VoIP is also a good thing for the telephony industry since usage of routers and cables which connect the Web together, reduces the monies needed to build the required infrastructure.

VoIP is making inroads—VoIP-enabled IP phones from ISPs are now more common, alongside the necessary services, while VoIP-enabled cell phones are expected soon. Consumer software such as Yahoo! Messenger and MSN Messenger allow voice chat amongst contacts which is essentially VoIP. Perhaps the biggest success story of VoIP software has been that of Skype—a P2P telephony network founded by Niklas Zennström and Janus Friis, the creators of KaZaA. On October 20, 2004, 1 million Skype users were online at once. The software allows Skype users to talk to each other for free; it can also call a land line for a small fee. Available for all leading platforms including Pocket PC handhelds, Skype has been widely proclaimed as the herald of cheaper voice communication.

But VoIP has its share of problems, the largest being that of voice quality—mainly that it is nothing great. Noisy connections and dropped packets (hence delayed or missed conversation threads) are expected and encountered. A possible reason for this could be patents over Internet Protocol Telephony. Experts point out that the field is rife with patents which cover almost all possible compression algorithms.

It is only a matter of time though before VoIP becomes the de facto means of voice communication, after all, it makes perfect economic sense.

